

LedgerWorks ERP

Complete Functional Specification Document (FSD)

Technical Reference & Development Roadmap

Version: Branch-5 Project Status: 95% Complete

1. PROJECT OVERVIEW

LedgerWorks ERP is a web-based ERP application developed using:

Frontend:

- React JS
- Axios
- React Router

Backend:

- Spring Boot
- Spring Data JPA
- REST APIs

Database:

- MySQL 8

Purpose: Provide a complete ERP solution for small and medium manufacturing and trading businesses.

Major Areas:

- Masters
 - Inventory
 - Sales
 - Purchase
 - Production
 - Accounting
 - Reporting
 - Backup & Recovery
 - Security
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2. SYSTEM ARCHITECTURE

Frontend (React) ↓ REST APIs ↓ Spring Boot ↓ Service Layer ↓ Repository Layer ↓ MySQL Database

Benefits:

- Separation of concerns
 - Easy maintenance
 - Easy deployment
 - Scalable architecture
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3. MASTER MODULES

Customer Master

Status: Completed

Purpose: Stores customer information used across the ERP.

Fields:

- Customer Name
- GST Number
- Address
- Mobile
- Email

Business Usage: Used in:

- Invoice
- Delivery Challan
- Outstanding Reports
- GST Reports

Implemented Features:

- Add Customer
- Update Customer
- Delete Customer
- Search Customer
- Customer Listing

Future Improvements:

- Customer Credit Limit
 - Customer Category
 - Customer Document Upload
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Vendor Master

Status: Completed

Purpose: Stores supplier information.

Business Usage: Used in Purchase module.

Implemented Features:

- Add Vendor
- Edit Vendor
- Delete Vendor
- Search Vendor

Future Improvements:

- Vendor Rating
 - Vendor Performance Analysis
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Item Master

Status: Completed

Purpose: Central inventory database.

Business Usage: Used in:

- Purchase
- Production
- Sales
- Stock Reports

Implemented Features:

- Add Item
- Edit Item
- Delete Item

- Search Item

Future Improvements:

- Barcode
 - QR Code
 - Item Images
 - Batch Tracking
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4. INVENTORY MANAGEMENT

Stock Increase

Status: Completed

Purpose: Adds inventory after purchase or production.

How It Works: Purchase Entry → Stock Ledger Update → Inventory Increase

Benefits: Accurate stock availability.

Stock Decrease

Status: Completed

Purpose: Reduces inventory after sales or consumption.

How It Works: Invoice/Production → Stock Ledger Update → Inventory Decrease

Benefits: Real-time stock tracking.

Low Stock Report

Status: Completed

Purpose: Identifies stock below minimum level.

Benefits: Prevents stock shortages.

Future Improvements:

- Automatic Purchase Suggestion
 - Reorder Level Alerts
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5. PURCHASE MODULE

Status: Completed

Purpose: Manages procurement process.

Implemented Features:

- Purchase Entry
- Vendor Selection
- Item Selection
- Stock Update

Business Flow: Vendor → Purchase Entry → Stock Increase

Future Improvements:

- Purchase Return
 - Purchase Approval Workflow
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6. SALES MODULE

Sales Invoice

Status: Completed

Purpose: Generates customer invoices.

Implemented Features:

- Create Invoice
- Invoice Listing
- Invoice Search
- Outstanding Calculation

Business Flow: Customer → Invoice → Stock Reduction → Accounting Entry

Future Improvements:

- PDF Invoice
 - Email Invoice
 - WhatsApp Invoice
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Invoice Outstanding

Status: Completed

Purpose: Tracks pending customer payments.

Benefits: Improves collection management.

Future Improvements:

- Automated Reminder System
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Invoice Payment

Status: Completed

Purpose: Records customer payments.

Benefits: Updates outstanding balance automatically.

Future Improvements:

- Online Payment Gateway
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7. PRODUCTION MODULE

Status: Completed

Purpose: Supports manufacturing operations.

Implemented Features:

- Production Entry
- Production List
- Material Issue

Business Flow: Raw Material → Production → Finished Goods

Benefits: Tracks manufacturing activities.

Future Improvements:

- BOM (Bill of Material)
 - Production Costing
 - Work Orders
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8. DELIVERY CHALLAN MODULE

Status: Completed

Purpose: Dispatch goods before invoicing.

Implemented Features:

- Challan Creation
- Challan Listing
- Customer Mapping

Benefits: Supports dispatch-first business model.

Future Improvements:

- E-Way Bill Integration
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9. ACCOUNTING MODULE

Status: Completed

Implemented Reports:

- Ledger Accounts
- Ledger Statement
- Trial Balance
- Profit & Loss
- Balance Sheet
- Cash Flow

Benefits: Provides complete financial visibility.

Future Improvements:

- Auto Journal Posting
 - Year End Closing
 - Cost Centers
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10. GST MODULE

Status: Completed

Implemented Features:

- GST Report
- GST Analytics

Benefits: Simplifies GST compliance.

Future Improvements:

- GSTR-1 Export
 - GSTR-3B Export
 - GST Reconciliation
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11. DASHBOARD MODULE

Status: Completed

Displayed Metrics:

- Total Sales
- Outstanding Amount
- Overdue Amount
- Overdue Count

Benefits: Single-screen business summary.

Future Improvements:

- Graphs
 - Monthly Trends
 - KPI Dashboard
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12. BACKUP MANAGEMENT MODULE

Status: Completed

Implemented Features:

- Manual Backup
- Startup Auto Backup
- Daily Scheduled Backup
- Backup History
- Download Backup
- Delete Backup
- Restore Backup
- Keep Last 30 Backups

Business Benefits:

- Disaster Recovery
- Data Protection
- Audit Support
- Migration Support

Future Improvements:

- Cloud Backup
 - Backup Encryption
 - Backup Verification
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13. USER MANAGEMENT MODULE

Status: Planned (Branch-6)

Features Planned:

- User Creation
- Password Management
- User Status
- Role Assignment

Benefits: Improves security and accountability.

14. ROLE BASED ACCESS CONTROL

Roles:

- ADMIN
- ACCOUNTANT
- USER
- VIEWER

Updated Rules:

ADMIN

- Full Access

ACCOUNTANT

- Full Access
- Cannot Delete Backup
- Cannot Restore Backup

USER

- Operational Activities Only

VIEWER

- Read Only Access
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15. SECURITY MODULE

Status: Planned

Branch-6 Features:

- Spring Security
 - JWT Authentication
 - Login Screen
 - Password Encryption
 - Session Management
 - API Protection
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16. AUDIT LOG MODULE

Status: Planned

Purpose: Track every business activity.

Examples:

- Customer Created
- Invoice Deleted
- Backup Restored
- User Login

Benefits: Complete traceability.

17. NOTIFICATION MODULE

Status: Planned

Features:

- Payment Reminder
 - Low Stock Alert
 - Backup Status Alert
 - GST Due Reminder
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18. CLOUD SERVICES

Status: Planned

Features:

- Google Drive Backup
- AWS S3 Backup
- Azure Storage Backup

Benefits: Protection against hardware failure.

19. MOBILE APP

Status: Future Phase

Features:

- Dashboard
- Sales Entry
- Stock Lookup
- Reports

Platform:

- Android
 - iOS
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20. BRANCH-WISE COMPLETION

Branch-1

- Initial Setup

Branch-2

- Customer & Vendor

Branch-3

- Sales & Purchase

Branch-4

- Production & Accounting

Branch-5

- Backup Management
- Scheduler
- Auto Backup
- Restore
- Download
- Delete
- Retention Policy
- Role Matrix Documentation

Branch-6 (Upcoming)

- Security
 - Login
 - User Management
 - JWT
 - Audit Logs
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CURRENT PROJECT STATUS

Backend Completion: 95%

Frontend Completion: 90%

Database Completion: 95%

Overall Project Completion: 95%

Production Readiness: Ready for Internal Usage

Pending Enterprise Features:

- Security
- Audit Logs
- Notifications
- Cloud Backup
- Mobile App
- Multi Company Support